

39. Vehicle Detection in a Video Frame

AIM

To detect vehicles in a video using OpenCV.

CODE

```
import cv2

video = cv2.VideoCapture(r"C:\Users\chait\Downloads\vehicle.mp4")

vehicle_cascade = cv2.CascadeClassifier("cars.xml")

while True:
    ret, frame = video.read()

    if not ret:
        break

    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    vehicles = vehicle_cascade.detectMultiScale(gray, 1.1, 3)

    for x, y, w, h in vehicles:
        cv2.rectangle(frame, (x, y), (x+w, y+h), (255, 0, 0), 2)

    cv2.imshow("Vehicle Detection", frame)

    if cv2.waitKey(30) & 0xFF == 27:
        break

video.release()
```

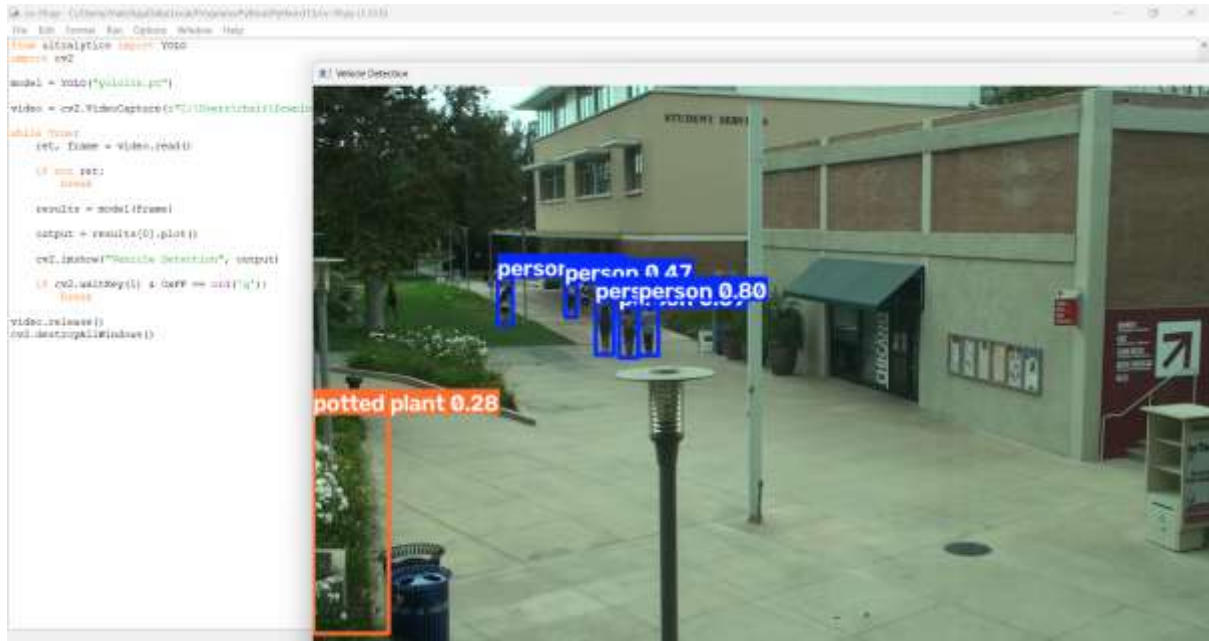
cv2.destroyAllWindows()

INPUT

A video containing vehicles and a vehicle Haar Cascade file named cars.xml.

OUTPUT

Rectangles are drawn around the detected vehicles.



RESULT

Thus, vehicles in the video were successfully detected using OpenCV.